

Dipan Banerjee

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Msc Digital Engineering

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GitHub Profile

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SUMMARY

Results-driven Data & Analytics Engineer with 3.5 years of experience building end-to-end data pipelines, machine learning models, and BI solutions across manufacturing, cybersecurity, and industrial domains. Hands-on expertise in Python, SQL, AWS, and Power BI, with a proven track record of automating processes, reducing costs (€20,000+ annual savings at Continental), and delivering real-time data-driven insights at scale. Currently pursuing MSc in Digital Engineering at OvGU Magdeburg, actively seeking full-time opportunities in Germany.

EDUCATION

•Master of Science in Digital Engineering

Otto-von-Guericke University Magdeburg, Germany

Grade: 2.3

Sept 2023 – Expected Dec 2026

•Bachelor of Technology in Electrical Engineering

Heritage Institute of Technology, Kolkata, India

Grade: 7.53/10

Aug 2016 – Aug 2020

PROFESSIONAL EXPERIENCE

•Data & Analytics Engineer (Working Student)

Continental Reifen AG - Global Process & Compounding Network

Oct 2025 - Present

Hannover-Lower Saxony, Germany

- Supporting the World Wide Compounding Network by coordinating manufacturing data processes and optimising global compound transfers through advanced analytics and automation.
- Orchestrated robust ETL workflows by extracting and transforming data via DBeaver from SQL and AWS Redshift environments, ensuring high data integrity and seamless reporting pipelines.
- Implemented machine learning models (Random Forest, Linear Regression) in Python to predict compound viscosity, evaluating performance using R^2 and RMSE metrics and embedding analytical outputs into Power BI reports.
- Performed statistical analysis and visualization (e.g., boxplots, correlation analysis) to assess compound variability and process stability, enabling early detection of deviations in mixing operations.
- Automated reporting processes using Excel Macros and SAP Analysis for Office (AFO) to streamline SAP data extraction, enhance reporting efficiency, and reduce manual intervention for production analytics.

•Cybersecurity Communications Intern

Continental Reifen AG

Feb 2025 - Aug 2025

Hannover-Lower Saxony, Germany

- Built and maintained a centralized web application using Flask, integrating security intelligence from 10+ cybersecurity tools (e.g., Zabbix, Landesk, CrowdStrike, Trellix, BMC Asset, VersionDog, Acronis, Active Directory), which empowered 21 globally distributed plants with unified threat visibility.
- Delivered automation initiatives that eliminated manual reporting and reconciliation efforts, generating an estimated €20,000+ in annual operational cost savings across global plant cybersecurity operations.
- Developed scalable ETL pipelines in C#, Python, and SQL, automating data ingestion and transformation from systems like MSSQL, streamlining analytics and response capabilities.
- Architected and automated Power BI dashboards for 21 global plants, integrating real-time cybersecurity data using advanced SQL, Power Query, and DAX, enabling actionable KPI insights and enabling dynamic reporting.
- Collaborated with global teams to enhance cybersecurity workflows, optimize automation strategies, and strengthen security policies across all plants.

•Data Analyst

Wipro Private Limited

Sept 2020 - Aug 2023

Kolkata, India

- Delivered advanced data analysis and visualization solutions, improving data-driven decision-making efficiency by 15% across enterprise operations.
- Collaborated closely with cross-functional teams, leveraging data-driven insights to develop and deploy innovative solutions.
- Leveraged SQL for optimized data management and retrieval, ensuring a robust foundation for operations.
- Developed and implemented Python-based automation frameworks for enterprise firewalls (Palo Alto, F5, Fortinet), decreasing security incidents by 40% and automating 80% of manual configuration workflows.
- Optimized API testing strategies by leveraging Postman, enhancing test coverage and mitigating potential risks.

MASTER THESIS

•Semantic-Retrieval-of-ICS-Security-of-BSI-and-NIST

Feb 2026 - Present

Python, Streamlit, LangChain, RAG, Supabase, NIST SP 800-82, BSI IT-Grundschutz, HITL Evaluation

- Built an OT cybersecurity analysis platform using RAG and Streamlit, integrating NIST SP 800-82 and BSI IT-Grundschutz for semantic retrieval and explainable requirement analysis.
- Developed semantic mapping, multi-LLM comparison, and LLM-as-a-Judge evaluation frameworks to assess technical accuracy, hallucination risk, and contextual relevance.
- Designed Human-in-the-Loop evaluation workflows and automated role/use-case generation modules, improving the practical applicability and trustworthiness of OT cybersecurity requirements.

RELEVANT PROJECTS

•Real-Time Fraud Detection & Investigation Platform

June 2026

Python, Apache Kafka, XGBoost, FAISS, Ollama, LangChain, Supabase, Streamlit, Docker

- Designed a real-time fraud detection platform combining XGBoost, rule-based engines, and Kafka streaming to analyze transaction behavior and generate consolidated fraud risk scores.
- Implemented a RAG-based investigation framework using FAISS, Sentence Transformers, and Ollama LLM to retrieve similar fraud cases and automatically generate contextual fraud investigation reports.

•Hybrid Retrieval-Augmented Generation (RAG) System for Intelligent Question Answering

April 2026

Python, Pinecone, Ollama

- Designed a scalable hybrid RAG architecture combining BM25 and vector similarity search (Pinecone) to balance precision and recall in information retrieval.

•Propaganda Detection in News Articles (AI Annotation)

July 2024 - Oct 2024

Python, Selenium Webdriver, BeautifulSoup, LLM(OpenAI GPT-4, Llama 3.0)

- Collected and manually annotated a diverse dataset of news articles (English and Hindi) for potential propaganda techniques.
- Automated data extraction processes using Selenium and BeautifulSoup, and employed LLM model(OpenAI GPT-4 and Llama 3.0) for large-scale Annotation.

•Transfer Learning on Time Series Analysis

April 2024 - Oct 2024

Python(Tensorflow, CNN)

- Applied transfer learning with CNN models to accurately predict CNC machine energy consumption, enhancing operational efficiency.

•Data Engineering Project: Automated Weather Data Pipeline

Aug 2024

Python, Apache Airflow, DAG, AWS(EC2,S3,Redshift), RestAPI

- Built a scalable ETL pipeline on AWS using Apache Airflow, automating the ingestion and loading of weather data into Redshift for real-time analytics.

ACHIEVEMENTS

Recognized for contributions to the project *Cybersecure Shopfloor through Automatic Reporting IPC Management Assessment 2025*, shortlisted in the **ContiTiresAward 2025 Local Jury Round** under the category *Our Opportunity*.

TECHNICAL SKILLS, AREA OF INTEREST AND RELEVANT COURSEWORK

Programming Languages: Python, SQL, C#(Basic), HTML

Development Tools: VS Code, Eclipse, Google Colab, Pycharm, Jupyter Notebook, Postman, Docker, Apache Airflow, Selenium Webdriver, dBeaver, Supabase

BI Tools: PowerBI

AI & LLM Technologies : LangChain, LangGraph, FAISS(Vector DB), LLAMA, Typesense

Version Control System: Git, Github, Gitlab

Microsoft-Office: Excel, Word, Powerpoint, Sharepoint

Databases: MySQL, PostgreSQL, Microsoft SQL, Redshift

Cloud: AWS(S3, Athena, Redshift, Glue, EC2)

Operation System: Linux, Windows

Relevant Coursework: Machine Learning, Visualization, Database Concepts, Software Engineering

Soft Skills: Self-learning, Presentation, Collaborative, Highly Motivated, TeamWork

LANGUAGE

English: C1

German: A2(Learning)